**3** Fig. 3 shows the time Lorraine spent in hours, $t$, answering e-mails during the working day. The data were collected over a number of months.

| Time in hours, $t$ | $0 \leqslant t < 1$ | $1 \leqslant t < 2$ | $2 \leqslant t < 3$ | $3 \leqslant t < 4$ | $4 \leqslant t < 6$ | $6 \leqslant t < 8$ |
|---|---|---|---|---|---|---|
| Number of days | 28 | 36 | 42 | 31 | 24 | 12 |

**Fig. 3**

**(a)** Calculate an estimate of the mean time per day that Lorraine spent answering e-mails over this period. **[1]**

**(b)** Explain why your answer to part **(a)** is an estimate. **[1]**

When Lorraine accepted her job, she was told that the mean time per day spent answering e-mails would not be more than 3 hours.

**(c)** Determine whether, according to the data in Fig. 3, it is possible that the mean time per day Lorraine spends answering e-mails is in fact more than 3 hours. **[1]**